

Improving relocalization accuracy via environment prior

Lars Hecker¹, Raman Besenfeller¹, David Farah¹, Fatemeh Sadat Daneshmand², Elisabetta Fedele¹, Linfei¹

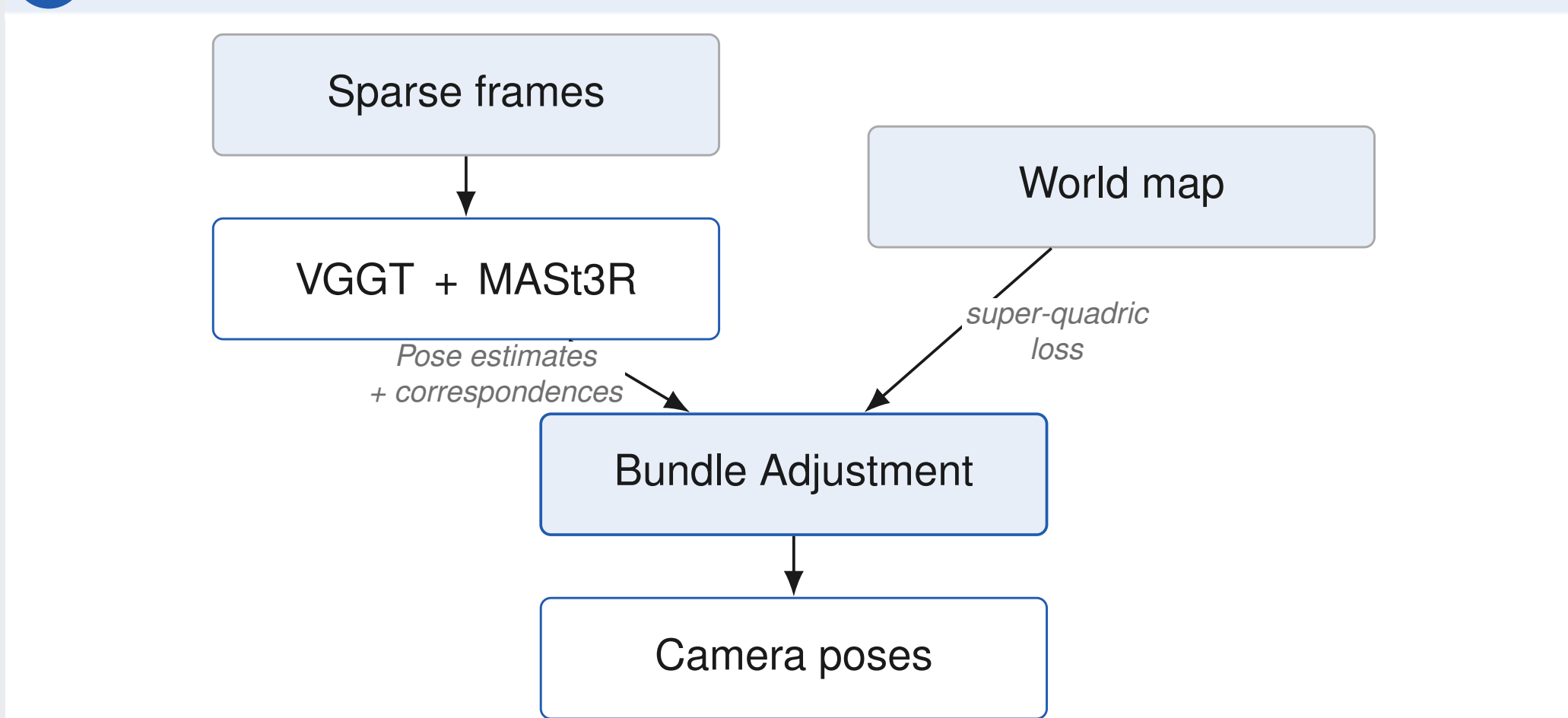
¹ETH Zurich ²ZHAW

On top of bundle adjustment, a super-quadric environment prior improves camera-pose accuracy most where views are sparse and barely overlap.
+1.5 AUC@5 at 6 views (up to +2.0 at 4) — but only +0.2 once 10 views already pin the geometry

1 The Problem

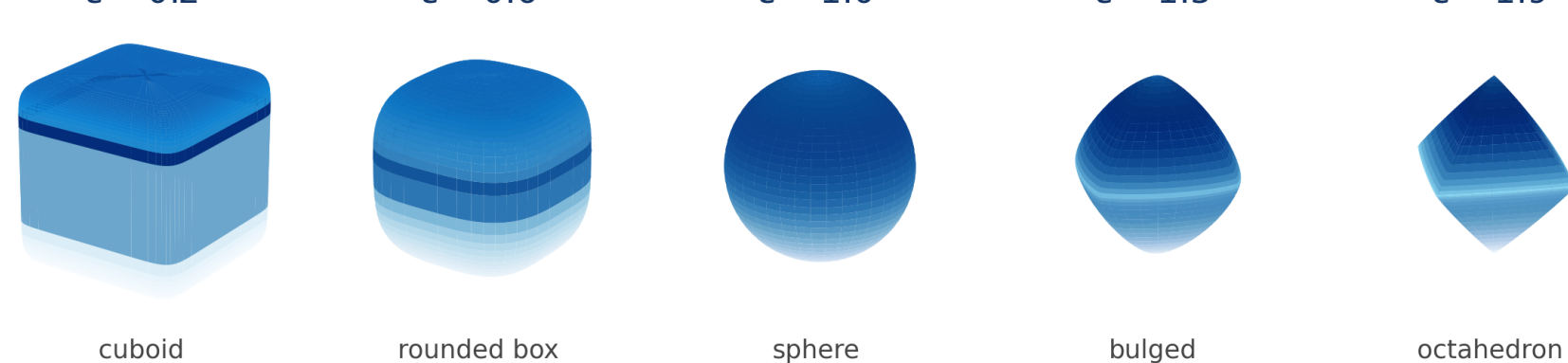
In relocalization (the “kidnapped robot”), camera pose must be recovered accurately *given* a known environment. Transformer-based SfM (e.g. VGGT [1]) gives fast, strong feed-forward poses but **cannot exploit such a prior** — and it degrades when views are few and barely overlap. We bridge this gap with a **super-quadric** scene prior [4].

2 Method Overview



3 The Environment Prior

A super-quadric is one shape whose roundness is set by an exponent ϵ — from box to sphere to octahedron



A super-quadric is a single shape family: two exponents set its roundness, three half-extents set its size, and a pose places it — so a handful tile a room.

SuperDec [4] fits a compact set of **super-quadrics** to the scene point cloud (right). During BA, each triangulated point is pulled onto its nearest primitive by a one-sided point-to-quadric term, re-associated EM-style every few iterations:

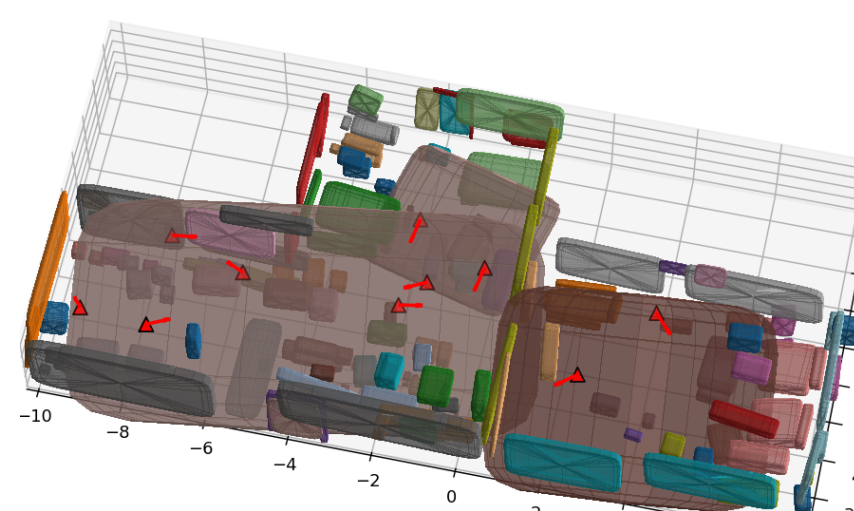


Fig. 1: a scene as 189 super-quadrics + camera poses.

$$E = \underbrace{\sum_{i,j} \rho(\|\pi(R_i, t_i, X_j) - x_{ij}\|^2)}_{\text{reprojection}} + \underbrace{\sum_j \left(\lambda \|q_j\| \cdot \max(0, 1 - F(q_j)^{-\epsilon_1}) \right)^2}_{\text{one-sided surface prior}}$$

where $F(\cdot)$ is the inside–outside function of the super-quadric nearest to point X_j and λ the prior weight [5].

4 Setup

- Dataset:** Aria Synthetic Environments (10 scenes)
- Models:** VGGT for feed-forward poses (via MapAnything [2]), MAST3R for matching [3]; SuperDec [4] for the prior
- Optimization:** full triangulation + surface BA in Ceres; **metric:** pose AUC@5° (higher is better)

5 Results

Three stages: raw VGGT poses, **bundle adjustment**, then BA + **our super-quadric prior**. BA does the heavy lifting (raw VGGT averages only 9.3 AUC@5 at 10 views, 20.4 at 6); the prior then adds a consistent gain *on top of* BA that grows as views get sparser.

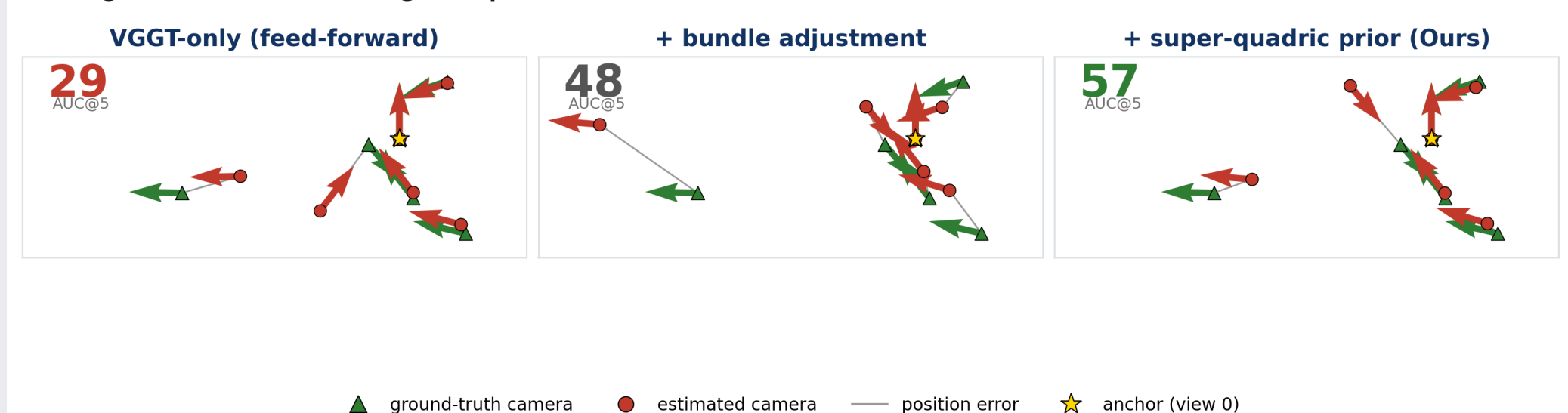
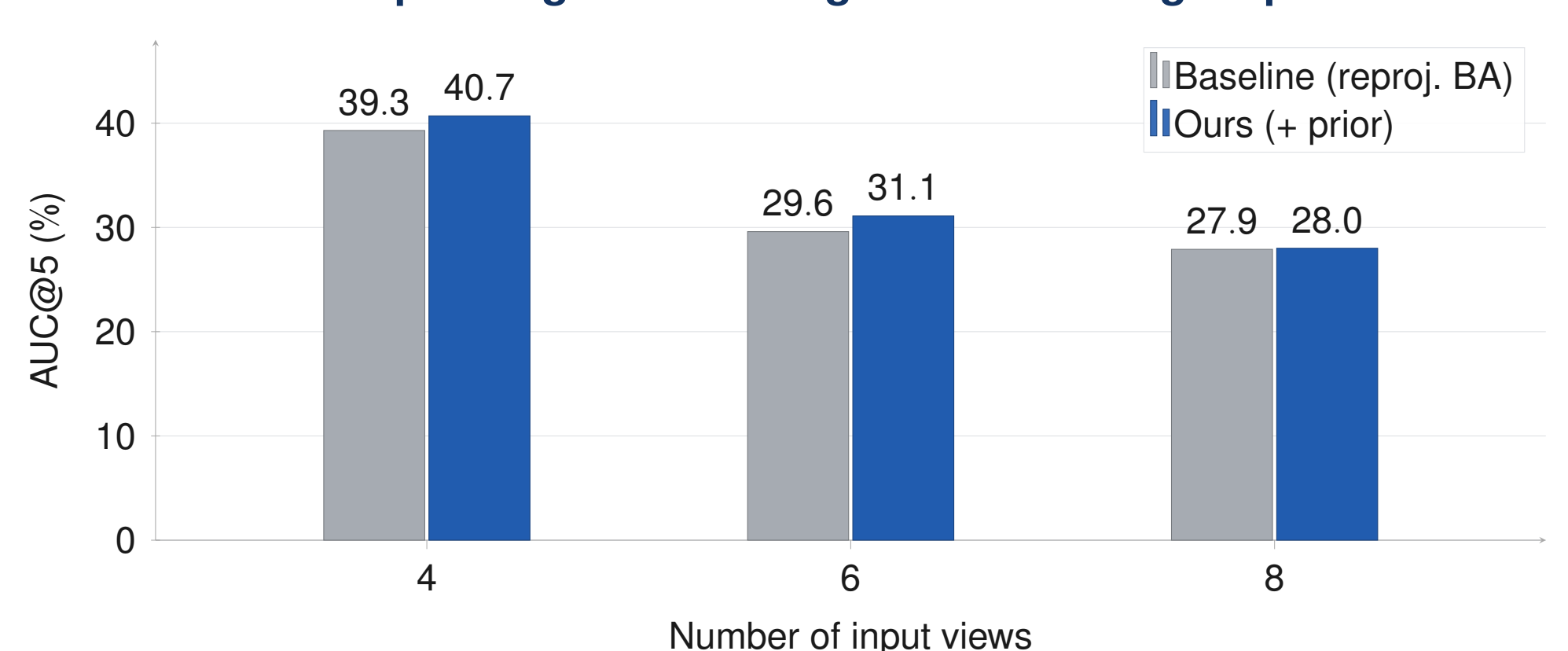


Figure 2: the pipeline on one scene (scene 6, 6 views) — the same 6 views at each stage. Bundle adjustment corrects raw VGGT; the super-quadric prior then pulls the remaining cameras onto ground truth (grey = position error). Pose AUC@5: 29 → 48 → 57.

The prior’s gain over BA grows as views get sparser



AUC@5 (%)	# input views			
	4	6	8	10
Baseline	39.3	29.6	27.9	29.4
Ours	40.7	31.1	28.0	29.6
Δ	+1.3	+1.5	+0.1	+0.2

Up to +2.0 at 4 views with light λ tuning. At 6 views the prior helps 3/10 scenes (up to +9.3) and hurts none.

6 Conclusion

At high view counts dense correspondences already constrain the poses, so the prior is largely redundant (+0.2 at 10 views). As views drop and co-visibility collapses, reprojection alone under-constrains the geometry — and the super-quadric surface fills exactly that gap, recovering up to +2.0 AUC@5 at 4 views. **Encoding an environment prior as a super-quadric cloud helps relocalization most where it matters: sparse, low-covisibility views.**

References & Code

- [1] Wang et al., VGGT: Visual Geometry Grounded Transformer. CVPR 2025.
 [2] Keetha et al., MapAnything: Universal Feed-Forward Metric 3D Reconstruction. 3DV 2026.
 [3] Leroy et al., Grounding Image Matching in 3D with MAST3R. ECCV 2024.
 [4] Fedele et al., SuperDec: 3D Scene Decomposition with Superquadric Primitives. ICCV 2025.
 [5] Mueller et al., Reconstructing People, Places, and Cameras. CVPR 2025.



Code & report